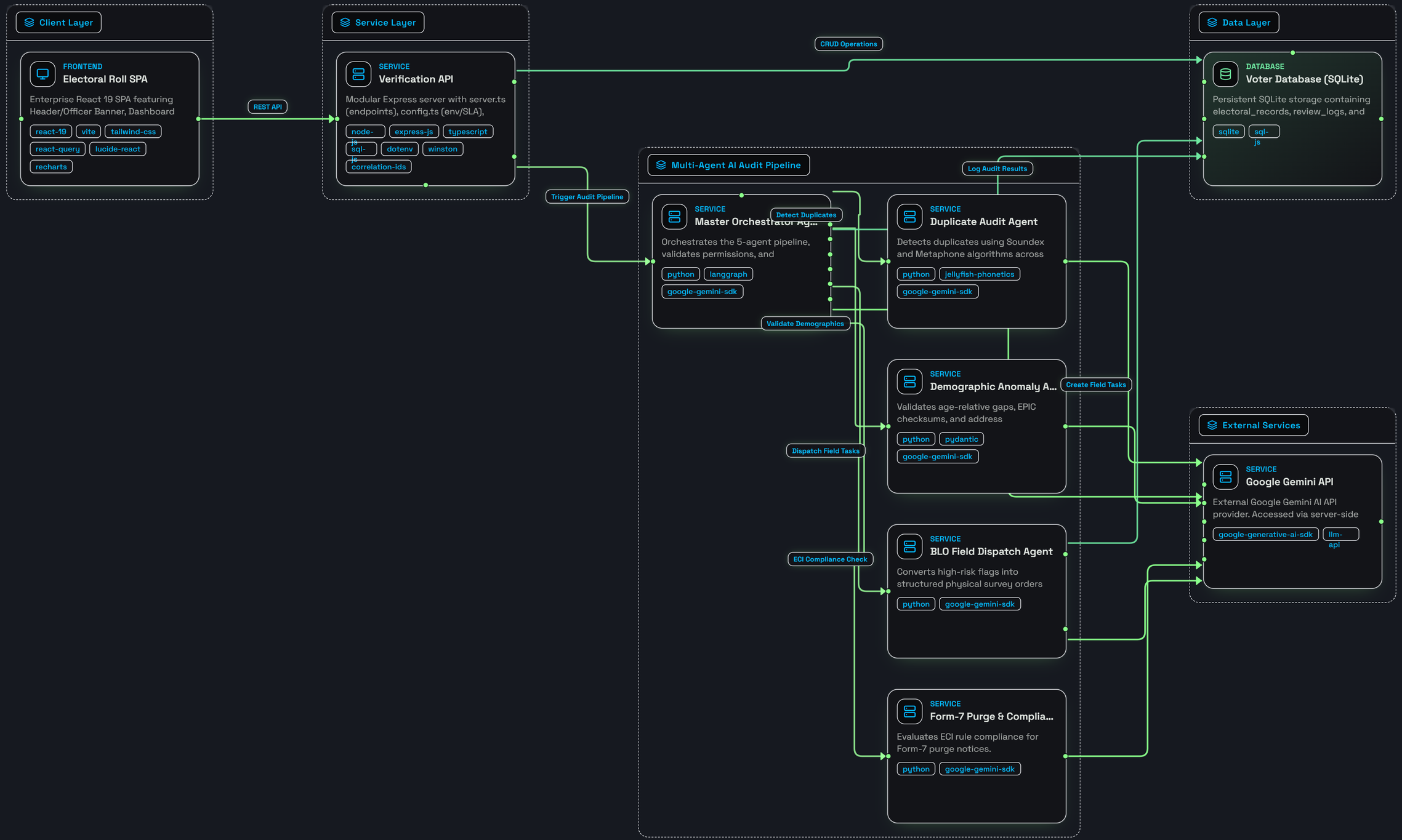
Client Layer
FRONTEND
Electoral Roll SPA
Enterprise React 19 SPA featuring Header/Officer Banner, Dashboard
react-19
vite
tailwind-css
react-query
lucide-react
recharts
REST API
Service Layer
SERVICE
Verification API
Modular Express server with server.ts (endpoints), config.ts (env/SLA),
node-js
express-js
typescript
sql-js
dotenv
winston
correlation-ids
CRUD Operations
Data Layer
DATABASE
Voter Database (SQLite)
Persistent SQLite storage containing electoral_records, review_logs, and
sqlite
sql-js
Trigger Audit Pipeline
Multi-Agent AI Audit Pipeline
Log Audit Results
SERVICE
Master Orchestrator Ag...
Orchestrates the 5-agent pipeline, validates permissions, and
python
langgraph
google-gemini-sdk
Detect Duplicates
SERVICE
Duplicate Audit Agent
Detects duplicates using Soundex and Metaphone algorithms across
python
jellyfish-phonetics
google-gemini-sdk
Validate Demographics
SERVICE
Demographic Anomaly A...
Validates age-relative gaps, EPIC checksums, and address
python
pydantic
google-gemini-sdk
Create Field Tasks
Dispatch Field Tasks
External Services
SERVICE
Google Gemini API
External Google Gemini AI API provider. Accessed via server-side
google-generative-ai-sdk
llm-api
ECI Compliance Check
SERVICE
BLO Field Dispatch Agent
Converts high-risk flags into structured physical survey orders
python
google-gemini-sdk
SERVICE
Form-7 Purge & Complia...
Evaluates ECI rule compliance for Form-7 purge notices.
python
google-gemini-sdk